

# Portfolio Temporal Structure Learning with Dynamic Bayesian Networks

*Category: Applied AI*

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## 1 Problem Statement

Financial portfolios contain hidden temporal dependencies: assets influence each other over time in patterns that traditional models don't capture explicitly. A covariance matrix tells you which assets move together, but not which assets *predict* which others.

This project learns to identify these predictive structures. We represent the portfolio as a directed graph where an edge from asset A to asset B means asset A's returns yesterday help predict asset B's returns today. We then compare sparse networks (with only significant relationships) against dense ones (using all relationships) to determine whether explicit structure learning improves out-of-sample predictions.

## 2 Representation: Dynamic Bayesian Networks

We model the portfolio as a directed acyclic graph (DAG) with assets as nodes and temporal edges representing predictive relationships. An edge  $A \rightarrow B$  at lag-1 means the previous period's return of A influences the current period's return of B. This representation has three key properties: it explicitly encodes temporal causality, it is acyclic (preventing circular causality across time), and it can be sparse (containing only significant relationships).

## 3 Learning: Score-Based Structure Learning

We learn the DAG structure by optimizing a penalized score function:

$$\hat{G} = \arg \min_{G \in \mathcal{G}} [BIC(G) + \lambda|E|]$$

where BIC measures goodness-of-fit,  $|E|$  is the number of edges, and acyclicity is guaranteed by the topological ordering of nodes.

For M1, we use a simplified approach: fit independent autoregressive models for each asset, extract coefficients, and keep only those significant at the  $p < 0.05$  level. This is less sophisticated than full score-based optimization but demonstrates the learning concept while remaining computationally simple for a proof-of-concept.

## 4 Inference: Temporal Prediction

Given a learned DAG structure, inference computes predictions from parent nodes and evaluates forecast accuracy on held-out test data. This directly measures whether the learned structure improves out-of-sample performance.

## 5 State-of-the-Art Reference

Our base paper is Pamfil et al. (2020), “DYNOTEARS: Structure Learning from Time-Series Data,” which introduces score-based learning of dynamic Bayesian networks with acyclicity constraints. The paper formulates DAG learning as continuous optimization over acyclic graphs and applies it to multivariate time-series. We adopt this framework applied to portfolio returns and simplify the implementation for M1 scope.

## 6 Baseline and Evaluation

The baseline is Vector Autoregression (VAR), which fits a full coefficient matrix without structure learning. This is the standard approach in finance. We compare using three metrics: (1) sparsity of the learned network (number of edges retained), (2) out-of-sample prediction error (RMSE and MAE), and (3) interpretability of the discovered edges.

## 7 Dataset

We use S&P 500 daily returns from 2015–2024 (2,500 trading days). The portfolio contains 10 major stocks: AAPL, MSFT, GOOGL, AMZN, TSLA (technology); JPM, GS (finance); XOM, CVX (energy); NVDA (semiconductors). We split temporally: train on 2015–2022, test on 2023–2024. This split respects the time-series nature of the data and tests generalization to recent market regimes.

## 8 Experimental Plan

For M1, we implement the data pipeline, fit ARX models per asset, extract significant edges, and compare against VAR. Subsequent milestones add full score-based optimization, regularization tuning, and multi-period analysis.

## 9 Team Responsibilities

Foram: data loading, VAR baseline implementation, prediction evaluation. Priya: ARX structure learning, edge extraction, DAG visualization and KPI computation.

## References

- [1] R. Pamfil et al., “DYNOTEARS: Structure Learning from Time-Series Data,” in *Proc. 23rd Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2020.
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